

QUESTION ONE (Start a new answer booklet)

Marks

- [2] (a) Evaluate $\int_0^3 \frac{dx}{9+x^2}$, giving your answer in exact form.
- [2] (b) Find the acute angle, in radians correct to two decimal places, between the lines $y = \frac{1}{3}x + 3$ and $y = -\frac{2}{3}x + 3$.
- [2] (c) Differentiate $y = \ln(\sin^{-1} x)$ with respect to x .
- [3] (d) Solve the inequality $x \geq \frac{1}{x}$.
- [3] (e) The polynomial equation $3x^3 - 2x^2 + 3x - 4 = 0$ has roots α , β and γ . Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$.

QUESTION TWO (Start a new answer booklet)

Marks

- [3] (a) Evaluate $\int_{-2}^0 \frac{x^2}{\sqrt{1-x^3}} dx$ using the substitution $u = 1 - x^3$.
- [3] (b) Prove the identity $\frac{\cos x - \cos 2x}{\sin 2x + \sin x} = \operatorname{cosec} x - \cot x$.
- [3] (c) Two parallel lines have equations $x - my + 1 = 0$ and $x - my - 1 = 0$.
- (i) Show that the point $(0, \frac{1}{m})$ lies on $x - my + 1 = 0$.
- (ii) Find m given that the perpendicular distance between the lines is 1 unit.
- [3] (d) The function $y = f(x)$ is defined by:
- $$f(x) = \begin{cases} \sin^{-1} x, & \text{for } -1 \leq x < 0, \\ \frac{\pi}{8}, & \text{for } x = 0, \\ \cos^{-1} x, & \text{for } 0 < x \leq 1. \end{cases}$$
- (i) Draw a neat sketch of $y = f(x)$, for $-1 \leq x \leq 1$.
- (ii) Evaluate $f(\frac{1}{2}) - f(-\frac{1}{2})$.

QUESTION THREE (Start a new answer booklet)

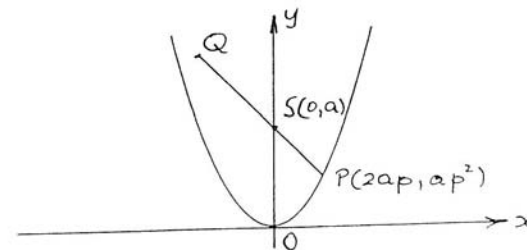
- [4] (a) (i) Given ${}^nC_r = \frac{n!}{r!(n-r)!}$ show that $\frac{r \times {}^nC_r}{{}^nC_{r-1}} = n - r + 1$.

(ii) Hence prove that:

$$\frac{{}^nC_1}{{}^nC_0} + \frac{2 \times {}^nC_2}{{}^nC_1} + \frac{3 \times {}^nC_3}{{}^nC_2} + \dots + \frac{n \times {}^nC_n}{{}^nC_{n-1}} = \frac{n}{2}(n+1).$$

- [4] (b) The coefficients of the terms in x^3 and x^{-3} in the expansion of $\left(mx + \frac{n}{x^2}\right)^6$ are equal, where m and n are non-zero real numbers. Prove that $\frac{m^2}{n^2} = \frac{10}{3}$.

- [4] (c)



In the diagram above, $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$ with focus $S(0, a)$. The point Q lies on PS produced, and Q divides PS externally so that $PQ:QS = -3:2$.

- (i) Prove that Q has coordinates $(-4ap, a(3-2p^2))$.
- (ii) Show that as P varies the locus of Q is another parabola. Find its equation, and write down the coordinates of its vertex and its focus in terms of a .

QUESTION FOUR (Start a new answer booklet)

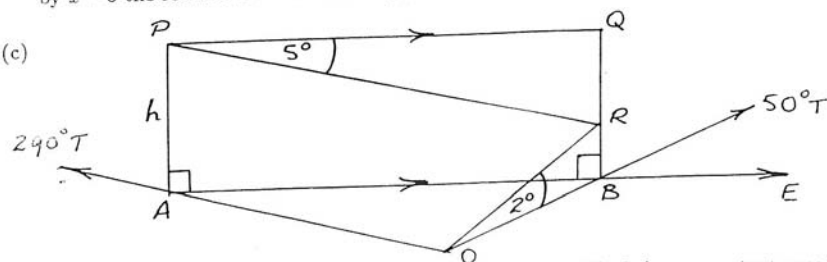
Marks

4 (a) (i) Prove that $\cos 2\theta = \frac{1-t^2}{1+t^2}$, where $t = \tan \theta$.

(ii) Hence show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$.

4 (b) $P(x)$ is an odd polynomial of degree 3. It has $x+4$ as a factor, and when it is divided by $x-3$ the remainder is 21. Find $P(x)$.

4 (c)



In the diagram above an aircraft is flying along the path PR . It has a constant speed of 300 km/hr and is descending at a steady angle of 5° . It flies directly over beacons at A and B , where B is due East of A . An observer at O first sights the aircraft over A at a bearing of $290^\circ T$. The observer sights the aircraft again 10 minutes later over B at a bearing of $50^\circ T$ and with an angle of elevation of 2° .

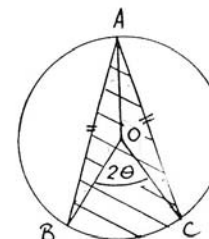
- Show that the aircraft has travelled 50 km in the 10 minutes between observations.
- Show that $\angle AOB = 120^\circ$.
- Prove that the observer at O is 19 670 metres, to the nearest 10 metres, from the beacon at B .
- Find the altitude h of the aircraft, to the nearest 10 metres, when it was originally sighted over A .

QUESTION FIVE (Start a new answer booklet)

3 (a) The radius of a spherical bubble is expanding at a constant rate of 0.02 mm/sec. Find the rate of increase of the volume when the radius is 20 mm.

3 (b) The displacement x metres of a particle at time t seconds moving in simple harmonic motion is $x = 4 \cos(\frac{\pi}{2}t + \epsilon)$, where $0 \leq \epsilon \leq 2\pi$. When $t = \frac{2}{3}$, $x = 2$ and $\dot{x} = \pi\sqrt{3}$ m/sec. Find the value of ϵ .

6 (c)



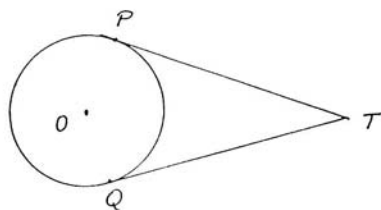
In the diagram above, AB and AC are equal chords of a circle with centre O and radius r . Let $\angle BOC = 2\theta$.

- The triangles ABO and ACO are congruent. Show that each has area $\frac{1}{2}r^2 \sin \theta$.
- Write down an expression in terms of r and θ for the area of the minor sector OBC .
- The shaded region, bounded by the two chords AB and AC and the minor arc BC , has area half that of the circle. Prove that $\theta + \sin \theta = \frac{\pi}{2}$.
- Taking $\theta = 0.8$ as a first approximation to the solution of $\theta + \sin \theta = \frac{\pi}{2}$, use one application of Newton's method to find a second approximation correct to two decimal places.

QUESTION SIX (Start a new answer booklet)

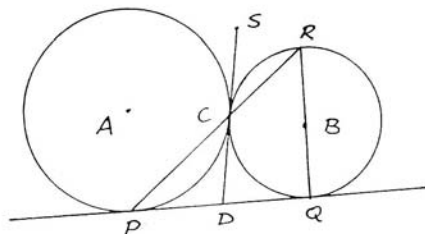
Marks

6 (a) (i)



In the diagram above, TP and TQ are tangents drawn from an external point T to a circle centre O . Prove that $TP = TQ$.

(ii)



In the figure above, two circles with centres A and B touch externally at point C . The common tangent at C intersects the direct common tangent PQ at D . The line QB produced meets the circumference of the smaller circle in R .

Copy the diagram into your answer booklet and prove that P , C and R are collinear.

6 (b) The rate of change of a quantity I with respect to t is given by $\frac{dI}{dt} = \frac{V}{L} - \frac{R}{L}I$, where R , L and V are constants.

(i) Show that $I = \frac{V}{R} + Ae^{-\frac{R}{L}t}$ satisfies this equation, where A is a constant.

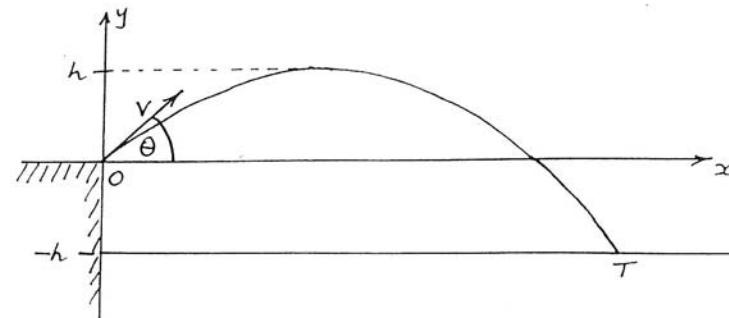
(ii) Deduce a relationship between I , V and R as t increases without bound.

(iii) Initially $I = 0$, and it is known that $V = 5$, $R = 2.2 \times 10^3$, and $L = 6.5 \times 10^{-4}$. Find the value of I , in scientific notation to 2 significant figures, when $t = 2 \times 10^{-4}$.

QUESTION SEVEN (Start a new answer booklet)

Marks

6 (a)



The diagram above shows the path of a projectile fired from the top O of a cliff. Its initial velocity is V m/s, its initial angle of elevation is θ and it rises to a maximum height h metres above O . It strikes a target T situated on a horizontal plane h metres below O .

The horizontal and vertical components of displacement in metres at time t seconds are given by $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$ respectively.

(i) Prove that $h = \frac{V^2 \sin^2 \theta}{2g}$.

(ii) Prove that the time taken for the projectile to reach its target is:

$$\frac{V \sin \theta (1 + \sqrt{2})}{g} \text{ seconds.}$$

(iii) Hence show that the distance from the target to the base of the cliff is:

$$\frac{V^2 (1 + \sqrt{2}) \sin 2\theta}{2g} \text{ metres.}$$

(b) (i) Show that $\sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4})$.

(ii) Prove that the derivative of $y = e^x \sin(x + \frac{\pi}{4})$ is given by $\frac{dy}{dx} = \sqrt{2} e^x \sin(x + \frac{\pi}{4})$.

(iii) Given the function $y = e^x \sin(x + \frac{\pi}{4})$, use mathematical induction that the n th derivative for positive integral n is:

$$\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin(x + \frac{n\pi}{4}).$$

QUESTION ONE

Marks

$$[2] \quad (a) \int_0^3 \frac{dx}{9+x^2} = \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_0^3 \quad \checkmark$$

$$= \frac{\pi}{12} \quad \checkmark$$

$$[2] \quad (b) \quad \tan \theta = \left| \frac{\frac{1}{3} + \frac{2}{3}}{1 - \frac{2}{9}} \right|$$

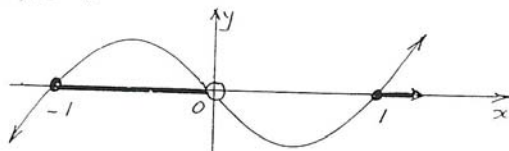
$$= \frac{9}{7} \quad \checkmark$$

So $\theta \doteq 0.91$ radians, to two decimal places. $\checkmark$

$$[2] \quad (c) \quad y = \ln(\sin^{-1} x)$$

$$\frac{dy}{dx} = \frac{1}{(\sin^{-1} x)\sqrt{1-x^2}} \quad \checkmark\checkmark$$

[3] (d)



$$x \geq \frac{1}{x}$$

$$x^3 \geq x \text{ and } x \neq 0$$

$$x(x-1)(x+1) \geq 0, \quad \checkmark$$

so $-1 \leq x < 0$ or $x \geq 1$. $\checkmark\checkmark$

$$[3] \quad (e) \quad 3x^3 - 2x^2 + 3x - 0 = 0,$$

$$\alpha + \beta + \gamma = \frac{2}{3},$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{1}{3}, \quad \checkmark \text{ for both}$$

$$\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} \quad \checkmark$$

$$= \frac{1}{2} \quad \checkmark$$

QUESTION TWO

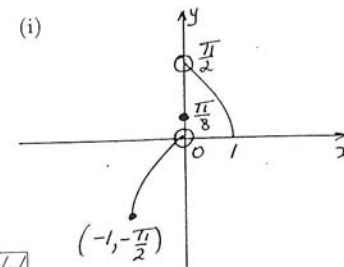
$$[3] \quad (a) \int_{-2}^0 \frac{x^2 dx}{\sqrt{1-x^2}} = -\frac{1}{3} \int_9^1 \frac{du}{u^{\frac{1}{2}}} \quad \checkmark$$

$$= -\frac{1}{3} \left[2u^{\frac{1}{2}} \right]_9^1 \quad \checkmark$$

$$= -\frac{2}{3} + \frac{2}{3} \times 3$$

$$= \frac{4}{3} \quad \checkmark$$

[3] (d) (i)



$$[3] \quad (b) \text{ LHS} = \frac{\cos x - \cos 2x}{\sin 2x - \cos x}$$

$$= \frac{\cos x - 2\cos^2 x + 1}{2\sin x \cos x + \sin x} \quad \checkmark$$

$$= \frac{-(2\cos^2 x - \cos x - 1)}{\sin x(2\cos x + 1)}$$

$$= \frac{-(2\cos x + 1)(\cos x - 1)}{\sin x(2\cos x + 1)}$$

$$= \frac{1 - \cos x}{\sin x} \quad \checkmark$$

$$= \operatorname{cosec} x - \cot x \quad \checkmark$$

$$= \text{RHS.}$$

 $\checkmark\checkmark$

$$(ii) \quad f\left(\frac{1}{2}\right) - f\left(-\frac{1}{2}\right) = \cos^{-1} \frac{1}{2} - \sin^{-1} \left(-\frac{1}{2}\right)$$

$$= \frac{\pi}{3} - \left(-\frac{\pi}{6}\right)$$

$$= \frac{\pi}{2} \quad \checkmark$$

$$[3] \quad (c) \quad (i) \text{ LHS} = x - my + 1$$

$$= 0 - \frac{1}{m} \times m + 1$$

$$= 0$$

$$= \text{RHS.} \quad \checkmark$$

So the point lies on the line.

$$(ii) \quad 1 = \left| \frac{0 - m \times \frac{1}{m} - 1}{\frac{1}{m^2} - 1} \right| \quad \checkmark$$

$$1 = \frac{2}{\sqrt{1+m^2}}$$

$$\sqrt{1+m^2} = 2$$

$$m^2 = 3$$

so $m = \sqrt{3}$ or $m = -\sqrt{3}$. $\checkmark$

QUESTION THREE

Marks

$$\boxed{4} \quad (a) \quad (i) \quad \frac{r \times {}^nC_r}{{}^nC_{r-1}} = \frac{r \times n!}{r!(n-r)!} \times \frac{(r-1)!(n-r+1)!}{n!}$$

$$= \frac{r(r-1)!(n-r+1)(n-r)!}{r(r-1)!(n-r)!} \quad \checkmark$$

$$= n-r+1. \quad \checkmark$$

$$(ii) \quad LHS = \frac{{}^nC_1}{{}^nC_0} + \frac{2 \times {}^nC_2}{{}^nC_3} + \frac{3 \times {}^nC_3}{{}^nC_2} + \dots + \frac{n \times {}^nC_n}{{}^nC_{n-1}}$$

$$= (n-1+1) + (n-2+1) + (n-3+1) + \dots + (n-n+1)$$

$$= n + (n-1) + (n-2) + \dots + 1 \quad \checkmark$$

$$= \frac{n}{2}(2n + (n-1)(-1))$$

$$= \frac{n}{2}(n+1) \quad \checkmark$$

$$= RHS.$$

$$\boxed{4} \quad (b) \quad u_{r+1} = {}^6C_r(mx)^{6-r}(nx^{-2})^r$$

$$= {}^6C_r m^{6-r} n^r x^{6-3r}. \quad \checkmark$$

$$\text{For term in } x^3: \quad 6-3r=3$$

$$r=1$$

$$\text{so} \quad u_2 = {}^6C_1 m^5 n x^3. \quad \checkmark$$

$$\text{For term in } x^{-3}: \quad 6-3r=-3$$

$$r=3$$

$$\text{so} \quad u_4 = {}^6C_3 m^3 n^3 x^{-3}. \quad \checkmark$$

$$\text{Now} \quad {}^6C_1 m^5 n = {}^6C_3 m^3 n^3$$

$$6m^5 n = 20m^3 n^3$$

$$\frac{m^2}{n^2} = \frac{10}{3}. \quad \checkmark$$

$$\boxed{4} \quad (c) \quad (i) \quad x = \frac{0-4ap}{3-2}$$

$$= -4ap$$

$$y = \frac{3a-2ap^2}{3-2}$$

$$= a(3-2p^2).$$

$$\text{So } Q \text{ is } (-4ap, a(3-2p^2)). \quad \checkmark$$

$$(ii) \quad x = -4ap$$

$$\text{so } p^2 = \frac{x^2}{16a}$$

$$y = a \left(3 - \frac{2x^2}{16a^2} \right)$$

$$= 3a - \frac{x^2}{8a}$$

$$\text{so } x^2 = -8a(y-3a).$$

This is another parabola.

Vertex: $V(0, 3a)$.Focal Length: $2a$.Focus: $S(0, a)$. $\checkmark \checkmark$

QUESTION FOUR

$$\boxed{4} \quad (a) \quad (i) \quad LHS = \cos 2\theta$$

$$= \cos^2 \theta - \sin^2 \theta$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} \quad \checkmark$$

$$= \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

$$= \frac{1-t^2}{1+t^2} \quad \checkmark$$

$$= RHS.$$

$$(ii) \quad \text{Let} \quad \theta = \frac{\pi}{8}.$$

$$\cos \frac{\pi}{4} = \frac{1-t^2}{1+t^2} \quad \checkmark$$

$$\frac{1}{\sqrt{2}} = \frac{1-t^2}{1+t^2}$$

$$t^2(1+\sqrt{2}) = \sqrt{2}-1$$

$$t^2 = \frac{\sqrt{2}-1}{\sqrt{2}+1}$$

$$= \frac{\sqrt{2}-1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$$

$$= (\sqrt{2}-1)^2$$

$$\text{so} \quad t = \sqrt{2}-1, \quad t > 0. \quad \checkmark$$

$$\boxed{4} \quad (b) \quad P(x) \text{ is odd, of degree 3, so } x \text{ and } x-4 \text{ are also factors.}$$

$$\text{Now } P(x) = ax(x+4)(x-4) \quad \checkmark$$

$$P(3) = 21 \quad \checkmark$$

$$21 = 3a \times 7 \times (-1).$$

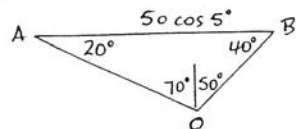
$$a = -1. \quad \checkmark$$

$$\text{Thus } P(x) = -x(x+4)(x-4)$$

$$= -x^3 + 16x. \quad \checkmark$$

(c) (i) $300 \text{ km/hr} = 50 \text{ km in 10 minutes.}$ ☒

(ii)

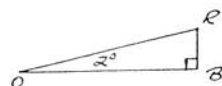
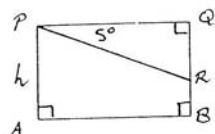


$$\begin{aligned}\angle AOB &= 70^\circ + 50^\circ \\ &= 120^\circ. \quad \checkmark\end{aligned}$$

(iii)

$$\begin{aligned}AB &= PQ \\ \cos 50^\circ &= \frac{PQ}{50} \\ PQ &= 50 \cos 5^\circ \\ \text{so } AB &= 50 \cos 5^\circ. \\ \text{Now } \frac{AB}{\sin 120^\circ} &= \frac{OB}{\sin 20^\circ} \\ OB &= \frac{AB \sin 20^\circ}{\sin 120^\circ} \\ &= \frac{50 \cos 5^\circ \sin 20^\circ}{\sin 120^\circ} \\ &\approx 19670 \text{ metres, to nearest 10 metres.} \quad \checkmark\end{aligned}$$

(iv)



$$\begin{aligned}\sin 5^\circ &= \frac{QR}{50000} \\ \text{so } QR &= 50000 \sin 5^\circ. \\ \tan 2^\circ &= \frac{RB}{19670} \\ \text{so } RB &= 19670 \tan 2^\circ. \\ \text{Now } h &= QR + RB \\ &\approx 5040 \text{ metres, to nearest 10 metres.} \quad \checkmark\end{aligned}$$

QUESTION FIVE

Marks

$$\begin{aligned}6 \quad (a) \quad (i) \quad \frac{dV}{dt} &= \frac{dV}{dr} \times \frac{dr}{dt} \quad \checkmark \\ &= 4\pi r^2 \times 0.02. \\ &= 4\pi \times 400 \times 0.02 \quad \checkmark \\ &= 32\pi \text{ mm}^3/\text{s.} \quad \checkmark\end{aligned}$$

(ii)

$$\begin{aligned}x &= 4 \cos\left(\frac{\pi}{2}t + \epsilon\right). \\ \text{When } t &= \frac{2}{3}, x = 2 \\ 2 &= 4 \cos\left(\frac{\pi}{3} + \epsilon\right) \\ \cos\left(\frac{\pi}{3} + \epsilon\right) &= \frac{1}{2}. \quad \checkmark \\ \text{Now } \dot{x} &= -2\pi \sin\left(\frac{\pi}{2}t + \epsilon\right). \\ \text{When } t &= \frac{2}{3}, \dot{x} = \pi\sqrt{3} \\ \pi\sqrt{3} &= -2\pi \sin\left(\frac{\pi}{3} + \epsilon\right) \\ \sin\left(\frac{\pi}{3} + \epsilon\right) &= -\frac{\sqrt{3}}{2}. \quad \checkmark \\ \text{Since } \cos\left(\frac{\pi}{3} + \epsilon\right) &> 0 \text{ and } \sin\left(\frac{\pi}{3} + \epsilon\right) < 0 \\ \frac{\pi}{3} + \epsilon &= \frac{5\pi}{3} \\ \epsilon &= \frac{4\pi}{3}. \quad \checkmark\end{aligned}$$

$$6 \quad (b) \quad (i) \quad \text{reflex } \angle BOC = 2\pi - 2\theta$$

$$\angle AOB = \angle AOC$$

$$= \frac{1}{2}(2\pi - 2\theta)$$

$$= \pi - \theta. \quad \checkmark$$

$$\text{Area} = \frac{1}{2}r^2 \sin(\pi - \theta)$$

$$= \frac{1}{2}r^2 \sin \theta. \quad \checkmark$$

$$(ii) \quad \text{Area} = \frac{1}{2}r^2 \times 2\theta$$

$$= r^2 \theta. \quad \checkmark$$

$$(iii) \quad 2 \times \frac{1}{2}r^2 \sin \theta + r^2 \theta = \frac{1}{2} \times \pi r^2$$

$$r^2 \sin \theta + r^2 \theta = \frac{\pi r^2}{2}$$

$$\theta + \sin \theta = \frac{\pi}{2}. \quad \checkmark$$

(iv) $\theta + \sin \theta = \frac{\pi}{2}$

$\theta + \sin \theta - \frac{\pi}{2} = 0$

Let $f(\theta) = \theta + \sin \theta - \frac{\pi}{2}$

$f'(\theta) = 1 + \cos \theta$

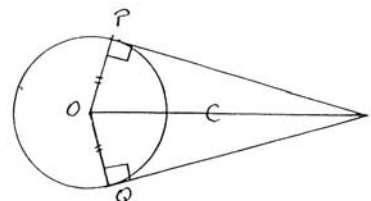
$\theta_2 = \theta_1 - \frac{f(\theta_1)}{f'(\theta_1)}$

$= 0.8 - \frac{0.8 + \sin 0.8 - \frac{\pi}{2}}{1 + \cos 0.8}$ ✓

≈ 0.83 correct to 2 decimal places. ✓

QUESTION SIX

6 (a) (i)



Join PO , OQ and OT .

In $\triangle OPT$ and $\triangle OQT$:

1. $PO = OQ$ (equal radii),

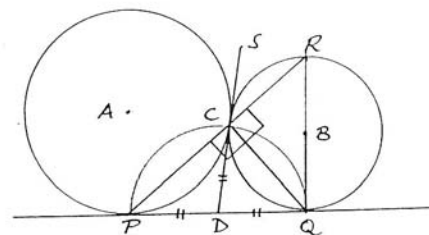
2. $\angle OPT = \angle OQT$ (radius perpendicular to tangent at point of contact),

3. $OT = OT$ (common side),

so $\triangle OPT \equiv \triangle OQT$ (RHS),

so $PT = QT$ (matching sides of congruent triangles). ✓✓

(ii)



Join CQ .

$PD = DC = DQ$ (tangents from an external point),
so circle with PQ as diameter passes through C . ✓

So $\angle PCQ = 90^\circ$ (angle in a semicircle, diameter PQ). ✓

also $\angle RCQ = 90^\circ$ (angle in a semicircle, diameter QR). ✓

so $\angle PCR = 180^\circ$,

So P , C , R are collinear. ✓

6 (b) (i) $I = \frac{V}{R} e^{-\frac{R}{L}t}$

$\frac{dI}{dt} = -\frac{R}{L} e^{-\frac{R}{L}t} \frac{V}{R}$ ✓

$= -\left(1 - \frac{V}{R}\right)$

$= -\frac{V}{L}$

$= -\frac{R}{L} I$ QED ✓

(ii) $\lim_{t \rightarrow \infty} I = \lim_{t \rightarrow \infty} \left(\frac{V}{R} + Ae^{-\frac{R}{L}t} \right)$

so $I = \frac{V}{R}$. ☒

(iii) When $t = 0$, $I = 0$,

so $0 = \frac{V}{R} + Ae^0$

so $A = -\frac{V}{R}$. ☒

Now $I = \frac{V}{R} - \frac{V}{R}e^{-\frac{R}{L}t}$

$= \frac{V}{R} \left(1 - e^{-\frac{R}{L}t} \right)$
 $= \frac{5}{2.2 \times 10^3} \left(1 - e^{-\frac{2.2 \times 10^3}{6.5 \times 10^{-3}} \times 2 \times 10^{-6}} \right)$ ☒

$\approx 1.1 \times 10^{-3}$ to two significant figures. ☒

QUESTION SEVEN

Marks

6 (a) (i) $y = Vt \sin \theta - \frac{1}{2}gt^2$

so $\dot{y} = V \sin \theta - gt$.

Let $\dot{y} = 0$,

so $t = \frac{V \sin \theta}{g}$. ☒

Now $y = V \sin \theta \left(\frac{V \sin \theta}{g} \right) - \frac{1}{2}g \left(\frac{V \sin \theta}{g} \right)^2$
 $= \frac{V^2 \sin^2 \theta}{g} - \frac{V^2 \sin^2 \theta}{2g}$
 $= \frac{V^2 \sin^2 \theta}{2g}$ ☒

(ii) Projectile reaches target where $y = -\frac{V^2 \sin^2 \theta}{2g}$.

so $-\frac{V^2 \sin^2 \theta}{2g} = Vt \sin \theta - \frac{1}{2}gt^2$ ☒

so $g^2t^2 - 2gVt \sin \theta - V^2 \sin^2 \theta = 0$

so $t = \frac{2gV \sin \theta \pm \sqrt{4g^2V^2 \sin^2 \theta + 4g^2V^2 \sin^2 \theta}}{2g^2}$
 $= \frac{2gV \sin \theta \pm 2\sqrt{2}gV \sin \theta}{g}$
 $= \frac{V \sin \theta(1 + \sqrt{2})}{g}$ since $t > 0$. ☒

(iii) $x = Vt \cos \theta$

$= V \cos \theta \times \frac{V \sin \theta(1 + \sqrt{2})}{g}$ ☒

$= \frac{V^2 \sin 2\theta(1 + \sqrt{2})}{2g}$. ☒

(b) (i) LHS $= \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$

$= \sqrt{2} \left(\sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} \right)$

$= \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x \right)$

$= \sin x + \cos x$

$= \text{RHS}$. ☒

(ii) $y = e^x \sin x$

$$\begin{aligned}\frac{dy}{dx} &= e^x \cos x + e^x \sin x \\ &= e^x (\sin x + \cos x) \\ &= \sqrt{2} e^x \sin \left(x + \frac{\pi}{4} \right). \quad \boxed{\checkmark}\end{aligned}$$

(iii) Prove $\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin \left(x + \frac{n\pi}{4} \right)$, for positive integral n .

$$\begin{aligned}\text{Step 1: Consider } n = 1, \text{ LHS} &= \frac{dy}{dx} \\ &= \sqrt{2} e^x \sin \left(x + \frac{\pi}{4} \right) \\ &= \text{RHS.} \quad \boxed{\checkmark}\end{aligned}$$

So the result is true for $n = 1$.

Step 2. Assume the result is true for positive integral k .

$$\text{So } \frac{d^k y}{dx^k} = (\sqrt{2})^k e^x \sin \left(x + \frac{k\pi}{4} \right). \quad \boxed{\checkmark}$$

We shall prove the result is true for $k + 1$,

$$\text{that is we prove, } \frac{d^{k+1} y}{dx^{k+1}} = (\sqrt{2})^{k+1} e^x \sin \left(x + \frac{(k+1)\pi}{4} \right).$$

$$\begin{aligned}\text{Now LHS} &= \frac{d^{k+1} y}{dx^{k+1}} \\ &= \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right) \\ &= \frac{d}{dx} \left((\sqrt{2})^k e^x \sin \left(x + \frac{k\pi}{4} \right) \right) \text{ by induction hypothesis } \quad \boxed{\checkmark} \\ &= (\sqrt{2})^k e^x \sin \left(x + \frac{k\pi}{4} \right) + (\sqrt{2})^k e^x \cos \left(x + \frac{k\pi}{4} \right) \\ &= (\sqrt{2})^k e^x \left(\sin \left(x + \frac{k\pi}{4} \right) + \cos \left(x + \frac{k\pi}{4} \right) \right) \\ &= (\sqrt{2})^k \times \sqrt{2} e^x \sin \left(x + \frac{k\pi}{4} + \frac{\pi}{4} \right) \\ &= (\sqrt{2})^{k+1} e^x \sin \left(x + \frac{(k+1)\pi}{4} \right). \quad \boxed{\checkmark \text{ with conclusion}} \\ &= \text{RHS.}\end{aligned}$$

Conclusion: It follows from steps 1 and 2 above by mathematical induction, the result is true for positive integral n .